Midwest Demographics Dataset

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Exploratory Analysis

INTRODUCTION

This report analyzes the Midwest dataset, a county-level demographic dataset derived from the 2000 U.S. Census. The dataset was selected because it includes 28 variables, combines categorical and continuous fields, and provides enough observations to support descriptive statistics, frequency tables, correlations, and multiple visualizations. It is a strong fit for an exploratory data analysis project because it captures population, education, poverty, race composition, metro status, and county classification in one file.

DATA SET DESCRIPTION

The dataset contains 437 county observations and 28 variables. Most variables are numeric percentage or count measures, while a smaller set of variables identify county, state, metro status, and county category. The dataset required very little repair because no missing values were found. The county field behaves more like an identifier than an analytical grouping variable, so the main categorical analysis focuses on state, metro status, and category.

Table 1: Data Types and Missing Data

Variable Name	Data Type	Pandas dtype	Percent Missing
PID	Nominal	int64	0.0
county	Nominal	object	0.0
state	Nominal	object	0.0
area	Ratio	float64	0.0
poptotal	Ratio	int64	0.0
popdensity	Ratio	float64	0.0
popwhite	Ratio	int64	0.0
popblack	Ratio	int64	0.0
popamerindian	Ratio	int64	0.0
popasian	Ratio	int64	0.0
popother	Ratio	int64	0.0
percwhite	Ratio	float64	0.0
percblack	Ratio	float64	0.0
percamerindan	Ratio	float64	0.0
percasian	Ratio	float64	0.0
percother	Ratio	float64	0.0
popadults	Ratio	int64	0.0
perchsd	Ratio	float64	0.0
percollege	Ratio	float64	0.0
percprof	Ratio	float64	0.0
poppovertyknown	Ratio	int64	0.0
percpovertyknown	Ratio	float64	0.0
percbelowpoverty	Ratio	float64	0.0
percchildbelowpovert	Ratio	float64	0.0
percadultpoverty	Ratio	float64	0.0
percelderlypoverty	Ratio	float64	0.0
inmetro	Nominal	int64	0.0
category	Nominal	object	0.0

SUMMARY STATISTICS

Table 2 summarizes the continuous variables in the dataset. The county population measures vary widely, which is expected because the dataset mixes rural and urban counties. Percentage-based variables show narrower spreads, but the poverty and education measures still reveal meaningful variation across counties.

Table 2: Summary Statistics for Midwest Dataset

Variable	Count	Mean	Std Dev	Min	25%	50%	75%	Max
PID	437.0	1437.339	876.39	561.0	670.0	1221.0	2059.0	3052.0
area	437.0	0.033	0.015	0.005	0.024	0.03	0.038	0.11
poptotal	437.0	96130.302	298170.54	1701.0	18840.0	35324.0	75651.0	5105067.0
popdensity	437.0	3097.743	7664.752	85.05	622.407	1156.208	2330.0	88018.397
popwhite	437.0	81839.915	200196.648	416.0	18630.0	34471.0	72968.0	3204947.0
popblack	437.0	11023.881	78958.267	0.0	29.0	201.0	1291.0	1317147.0
popamerindian	437.0	343.11	868.927	4.0	44.0	94.0	288.0	10289.0
popasian	437.0	1310.465	9518.394	0.0	35.0	102.0	401.0	188565.0
popother	437.0	1612.931	18526.541	0.0	20.0	66.0	345.0	384119.0
percwhite	437.0	95.558	7.087	10.694	94.886	98.033	99.075	99.823
percblack	437.0	2.676	5.136	0.0	0.116	0.539	2.601	40.21
percamerindan	437.0	0.799	4.536	0.056	0.158	0.215	0.384	89.177
percasian	437.0	0.487	0.628	0.0	0.174	0.297	0.521	5.07
percother	437.0	0.479	0.838	0.0	0.091	0.178	0.481	7.524
popadults	437.0	60972.613	191644.862	1287.0	12271.0	22188.0	47541.0	3291995.0
perchsd	437.0	73.966	5.843	46.912	71.325	74.247	77.195	88.899
percollege	437.0	18.273	6.262	7.336	14.114	16.798	20.55	48.079
percprof	437.0	4.447	2.408	0.52	2.998	3.814	4.949	20.791
poppovertyknown	437.0	93642.284	293235.058	1696.0	18364.0	33788.0	72840.0	5023523.0
percpovertyknown	437.0	97.11	2.75	80.902	96.895	98.17	98.599	99.86
percbelowpoverty	437.0	12.511	5.15	2.18	9.199	11.822	15.133	48.691
percchildbelowpovert	437.0	16.447	7.229	1.919	11.624	15.27	20.352	64.308
percadultpoverty	437.0	10.919	5.109	1.939	7.668	10.008	13.182	43.312
percelderlypoverty	437.0	11.389	3.661	3.547	8.912	10.869	13.412	31.162
inmetro	437.0	0.343	0.475	0.0	0.0	0.0	1.0	1.0

CATEGORICAL VARIABLES

The main categorical variables are state, metro status, and county category. Frequency tables and proportions show how the dataset is distributed across these groups. Illinois contributes the largest share of counties in the sample, about 34.32% of counties are metro counties, and the most common county category is AAR.

Table: Frequency and Proportion for state

state	Frequency	Proportion
IL	102	0.2334
IN	92	0.2105
OH	88	0.2014
MI	83	0.1899
WI	72	0.1648

Table: Frequency and Proportion for inmetro

inmetro	Frequency	Proportion
0.0	287.0	0.6568
1.0	150.0	0.3432

Table: Frequency and Proportion for category

category	Frequency	Proportion
AAR	193	0.4416
AAU	77	0.1762
LAR	30	0.0686
LHR	28	0.0641
HLU	26	0.0595
ALU	20	0.0458
HAU	20	0.0458
AHR	16	0.0366
ALR	11	0.0252
HAR	6	0.0137
LAU	3	0.0069
HLR	2	0.0046
LHU	2	0.0046
HHR	1	0.0023
HHU	1	0.0023
AHU	1	0.0023

CORRELATION ANALYSIS

The correlation matrix was computed using the continuous variables only. Some strong positive correlations are expected, especially between absolute population counts that scale together. One of the strongest negative relationships appears between percent white and percent black, which makes sense because both are parts of the same population composition.

Table 4: Correlation Table

Variable	PID	area	poptotal	popdensity	popwhite	popblack	popamerindian	popasian	popother	percwhite	percblack	percamerindian	percasian	percother	popadults	perchsd	percollege	percprof	poppovertyknown	percpovertyknown	percbelowpovert y	perccchildbelowpovert	percadultpovertty	percelderlypovertty	inmetro

PID	1.0	0.309	-0.009	0.02	0.005	-0.03	0.109	-0.041	-0.05	-0.009	-0.102	0.141	0.02	-0.074	-0.009	0.096	0.071	-0.108	-0.009	0.069	0.014	0.037	0.018	-0.104	0.006
area	0.309	1.0	0.067	-0.059	0.067	0.053	0.188	0.081	0.077	-0.047	-0.046	0.107	0.119	0.017	0.066	0.147	0.15	0.025	0.066	-0.114	0.073	0.052	0.102	-0.008	-0.106
poptotal	-0.009	0.067	1.0	0.898	0.989	0.947	0.815	0.923	0.858	-0.475	0.553	-0.031	0.451	0.462	1.0	0.137	0.301	0.276	1.0	0.035	-0.029	0.026	-0.031	-0.149	0.308
popdensity	0.02	-0.059	0.898	1.0	0.918	0.823	0.797	0.726	0.614	-0.521	0.631	-0.041	0.467	0.409	0.897	0.172	0.35	0.319	0.897	0.05	-0.052	0.021	-0.056	-0.202	0.388
popwhite	0.005	0.067	0.989	0.918	1.0	0.894	0.811	0.895	0.806	-0.481	0.562	-0.04	0.502	0.464	0.989	0.191	0.365	0.33	0.989	0.034	-0.065	-0.008	-0.061	-0.202	0.367
popblack	-0.03	0.053	0.947	0.823	0.894	1.0	0.785	0.864	0.838	-0.461	0.548	-0.016	0.303	0.398	0.946	0.018	0.152	0.151	0.948	0.041	0.055	0.107	0.038	-0.031	0.185
popamerindian	0.109	0.188	0.815	0.797	0.811	0.785	1.0	0.684	0.614	-0.613	0.517	0.253	0.382	0.361	0.812	0.141	0.283	0.225	0.815	-0.003	0.097	0.15	0.092	-0.116	0.265
popasian	-0.041	0.081	0.923	0.726	0.895	0.864	0.684	1.0	0.963	-0.332	0.345	-0.016	0.444	0.447	0.925	0.091	0.237	0.232	0.923	0.011	-0.013	0.013	-0.01	-0.083	0.173
popother	-0.05	0.077	0.858	0.614	0.806	0.838	0.614	0.963	1.0	-0.284	0.287	-0.01	0.293	0.482	0.86	0.02	0.117	0.114	0.858	0.026	0.007	0.038	-0.001	-0.036	0.111
percwhite	-0.009	-0.047	-0.475	-0.521	-0.481	-0.461	-0.613	-0.332	-0.284	1.0	-0.76	-0.595	-0.319	-0.342	-0.471	-0.019	-0.215	-0.221	-0.474	0.125	-0.377	-0.413	-0.35	-0.043	-0.264
percblack	-0.102	-0.046	0.553	0.631	0.562	0.548	0.517	0.345	0.287	-0.76	1.0	-0.047	0.322	0.309	0.549	0.014	0.237	0.257	0.552	-0.136	0.218	0.287	0.188	0.041	0.346
percamerindan	0.141	0.107	-0.031	-0.041	-0.04	-0.016	0.253	-0.016	-0.01	-0.595	-0.047	1.0	-0.062	-0.044	-0.031	-0.092	-0.082	-0.091	-0.031	0.008	0.374	0.36	0.346	0.117	-0.081
percasian	0.02	0.119	0.451	0.467	0.502	0.303	0.382	0.444	0.293	-0.319	0.322	-0.062	1.0	0.309	0.443	0.474	0.752	0.793	0.446	-0.355	-0.044	-0.148	0.08	-0.322	0.39
percother	-0.074	0.017	0.462	0.409	0.464	0.398	0.361	0.447	0.482	-0.342	0.309	-0.044	0.309	1.0	0.458	0.214	0.242	0.197	0.461	-0.002	-0.143	-0.105	-0.129	-0.276	0.259
popadults	-0.009	0.066	1.0	0.897	0.989	0.946	0.812	0.925	0.86	-0.471	0.549	-0.031	0.443	0.458	1.0	0.133	0.293	0.268	1.0	0.041	-0.03	0.027	-0.034	-0.146	0.304
perchsd	0.096	0.147	0.137	0.172	0.191	0.018	0.141	0.091	0.02	-0.019	0.014	-0.092	0.474	0.214	0.133	1.0	0.777	0.63	0.135	-0.122	-0.587	-0.624	-0.468	-0.75	0.447
percollege	0.071	0.15	0.301	0.35	0.365	0.152	0.283	0.237	0.117	-0.215	0.237	-0.082	0.752	0.242	0.293	0.777	1.0	0.8887	0.297	-0.266	-0.281	-0.363	-0.149	-0.526	0.4844
perc	-	0	0.	0.	0.	0.	0.2	0.	0.	-	0.	-	0.	0.	0.	0.	0.	1.	0.27	-	-	-	-	-	0.

prof	0.108	.025	276	319	33	151	25	232	114	0.221	257	0.091	793	197	268	63	887	0	1	0.403	0.148	0.276	0.001	0.403	438
poppovertyknown	-0.009	0.066	1.0	0.897	0.989	0.948	0.815	0.923	0.858	-0.474	0.552	-0.031	0.446	0.461	1.0	0.135	0.297	0.271	1.0	0.041	-0.029	0.026	-0.033	-0.148	0.306
percpovertyknown	0.069	-0.114	0.035	0.05	0.034	0.041	-0.003	0.011	0.026	0.125	-0.136	0.008	-0.355	-0.002	0.041	-0.122	-0.266	-0.403	0.041	1.0	-0.179	-0.016	-0.294	-0.005	0.01
percbelowpoverty	0.014	0.073	-0.029	-0.052	-0.065	0.055	0.097	-0.013	0.007	-0.377	0.218	0.374	-0.044	-0.143	-0.03	-0.587	-0.281	-0.148	-0.029	-0.179	1.0	0.95	0.974	0.751	-0.311
perchildbelowpovert	0.037	0.052	0.026	0.021	-0.008	0.107	0.15	0.013	0.038	-0.413	0.287	0.36	-0.148	-0.105	0.027	-0.624	-0.363	-0.276	0.026	-0.016	0.95	1.0	0.878	0.686	-0.27
percadultpoverty	0.018	0.102	-0.031	-0.056	-0.061	0.038	0.092	-0.01	-0.001	-0.35	0.188	0.346	0.08	-0.129	-0.034	-0.468	-0.149	-0.001	-0.033	-0.294	0.974	0.878	1.0	0.653	-0.27
percelderlypoverty	-0.104	-0.008	-0.149	-0.202	-0.202	-0.031	-0.116	-0.083	-0.036	-0.043	0.041	0.117	-0.322	-0.276	-0.146	-0.75	-0.526	-0.403	-0.148	-0.005	0.751	0.686	0.653	1.0	-0.477
inmetro	0.006	-0.106	0.308	0.388	0.367	0.185	0.265	0.173	0.111	-0.264	0.346	-0.081	0.39	0.259	0.304	0.447	0.484	0.438	0.306	0.01	-0.311	-0.27	-0.27	-0.477	1.0

DATA SET GRAPHICAL EXPLORATION

The figures below highlight the shape of important variables and the relationships among selected demographic measures. Each graph is labeled and discussed briefly in the surrounding narrative.

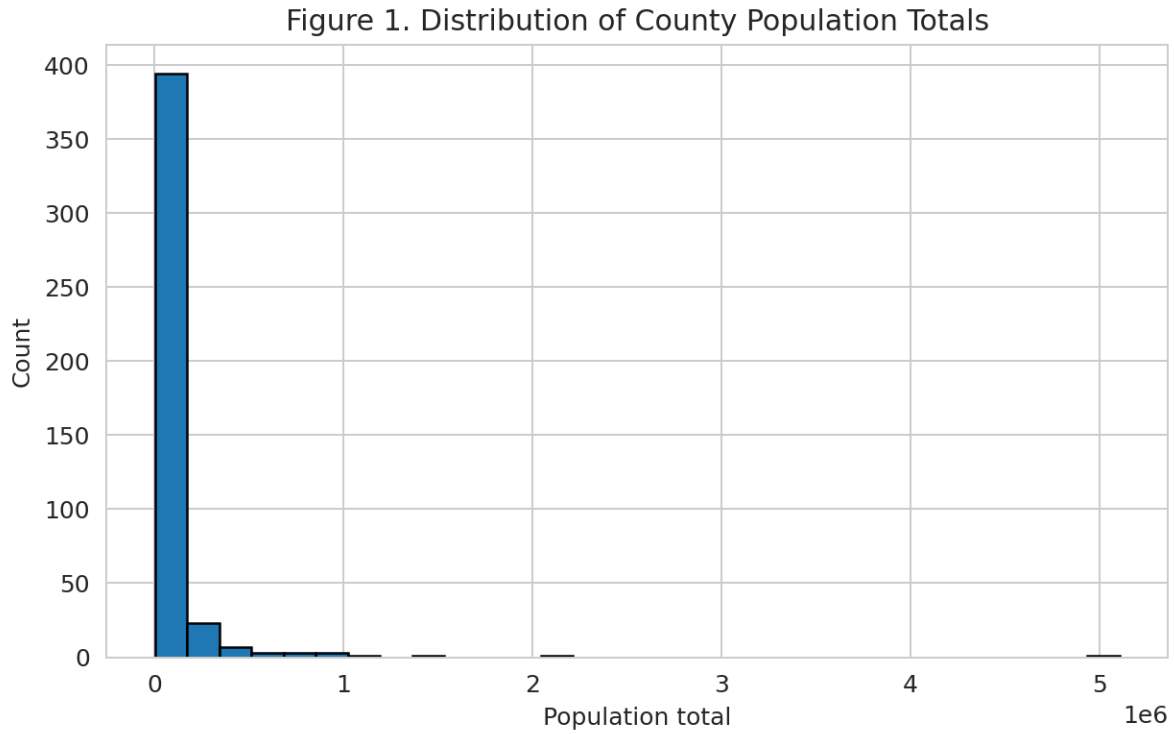

Figure 1: Distribution of county population totals. The distribution is strongly right-skewed.

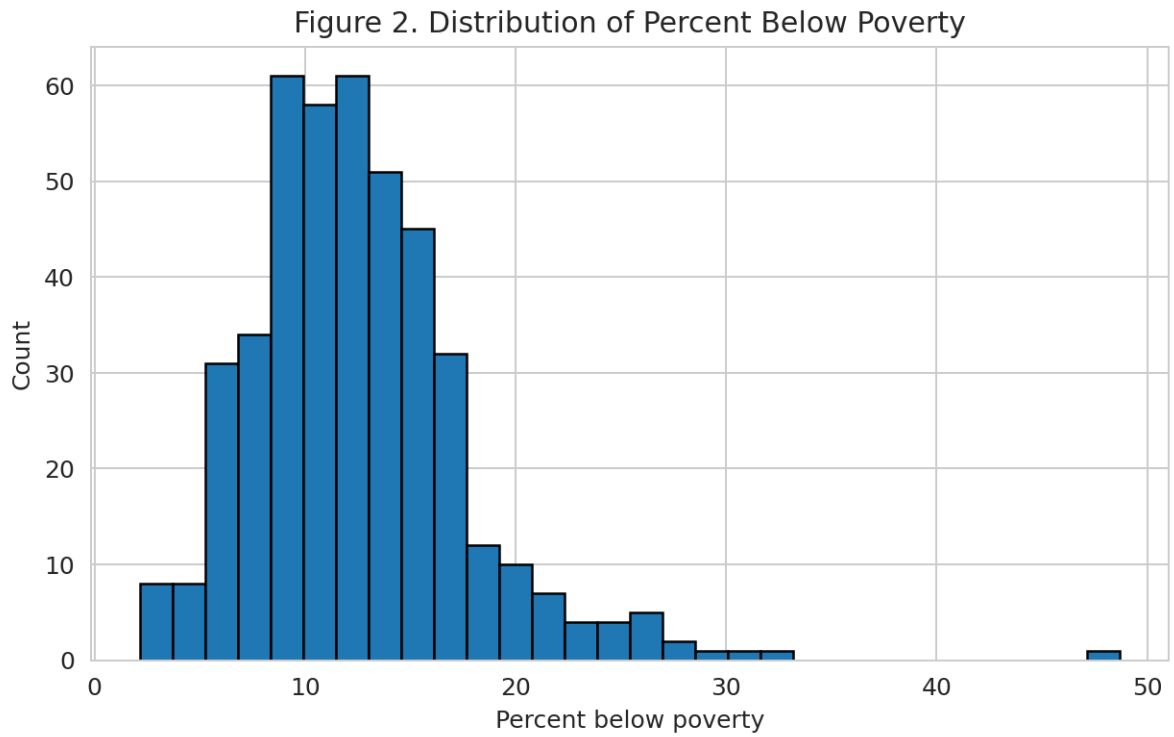

Figure 2: Distribution of percent below poverty. Most counties cluster in the low-to-mid range.

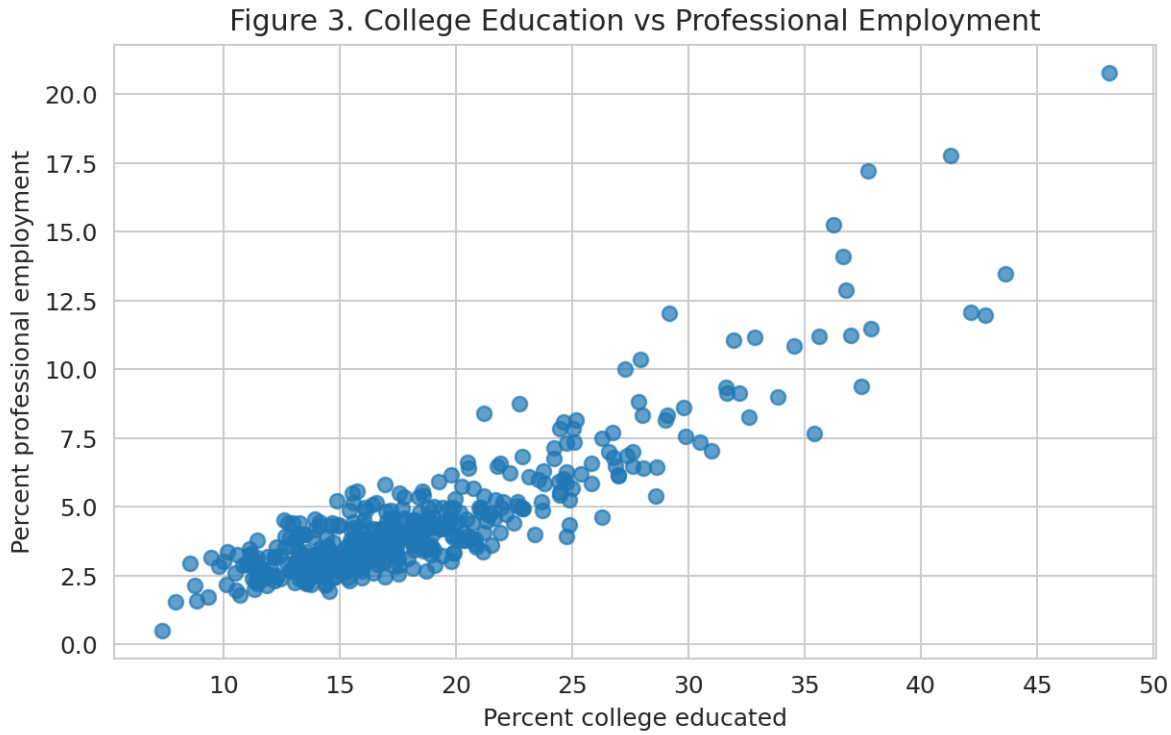

Figure 3: Percent college educated versus percent professional employment. The positive trend is visible.

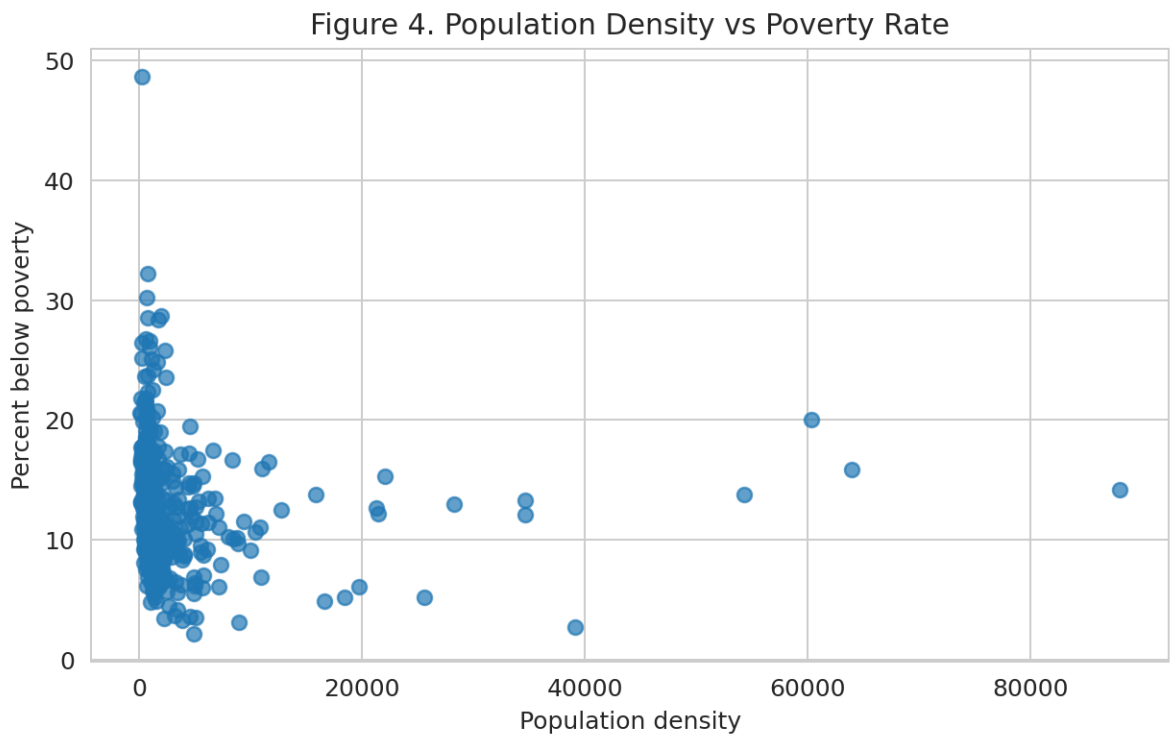

Figure 4: Population density versus poverty rate. The relationship is present but more scattered.

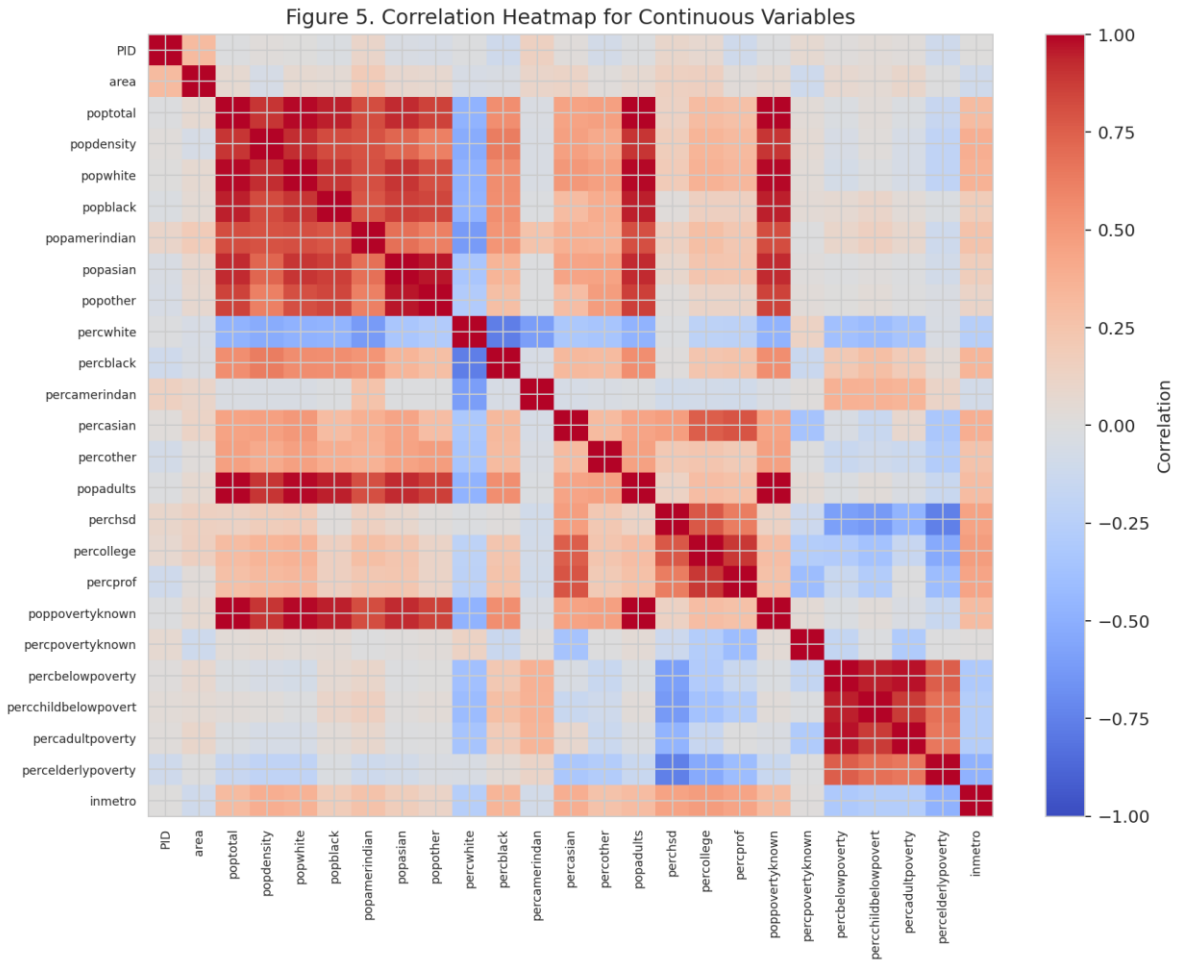

Figure 5: Correlation heatmap for continuous variables.

Figure 6. Pairwise Relationships Among Selected Continuous Variables

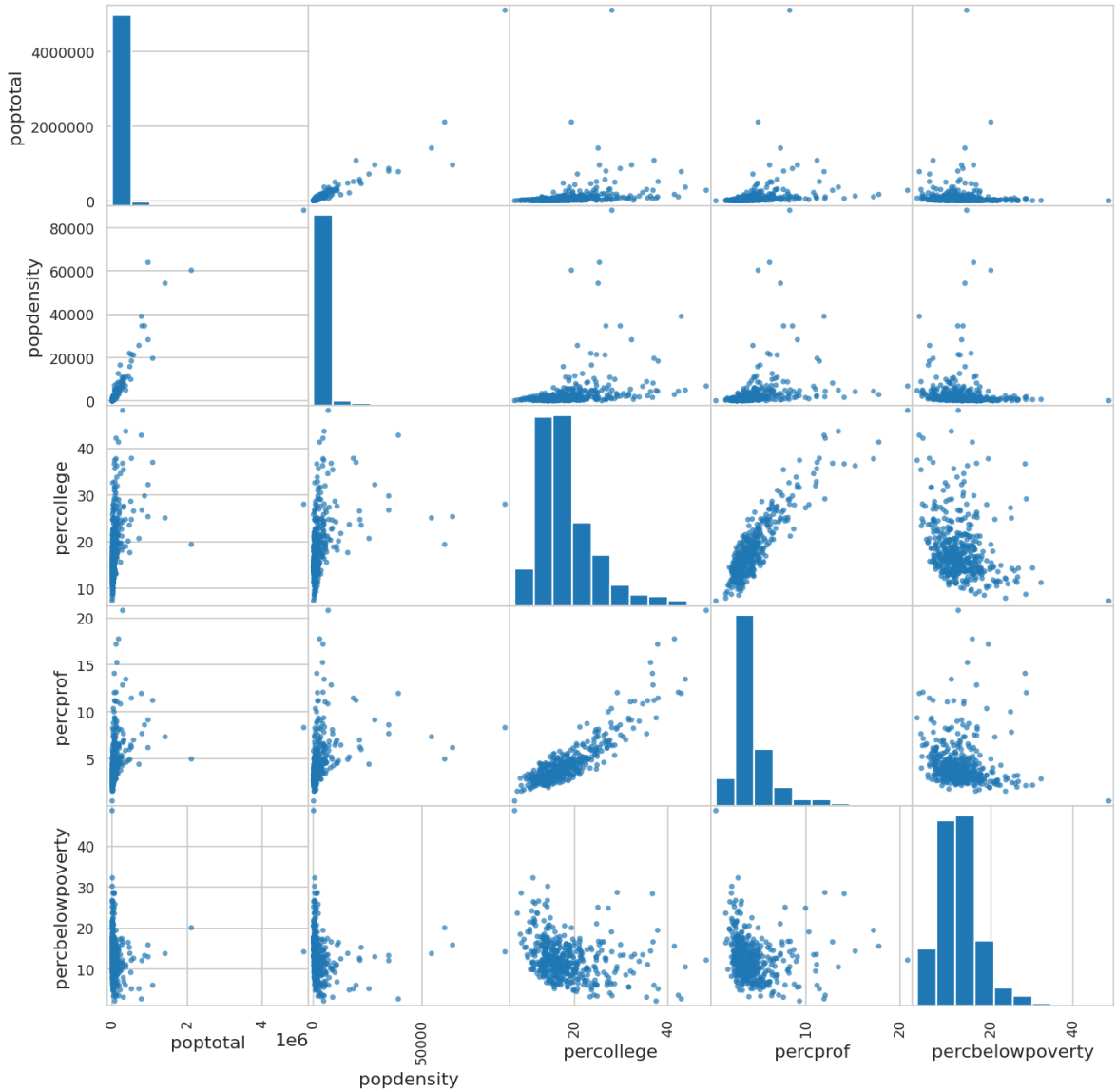

Figure 6: Pairwise plot for selected continuous variables.

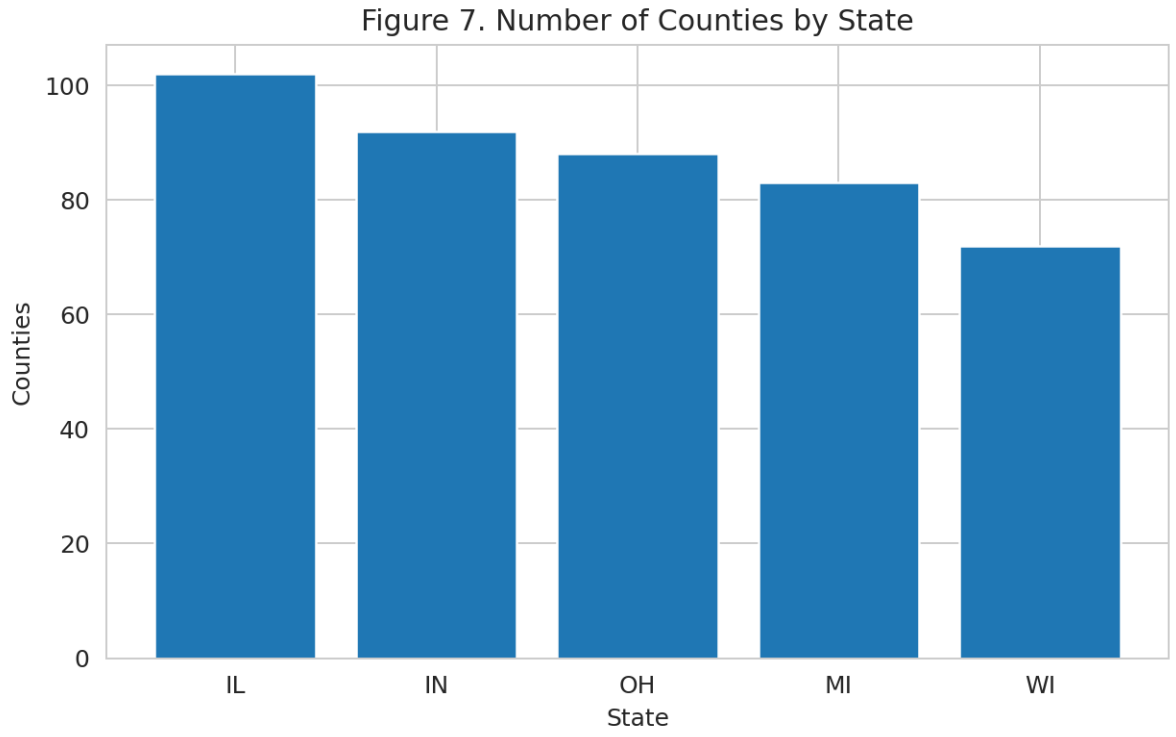

Figure 7: Number of counties by state in the dataset.

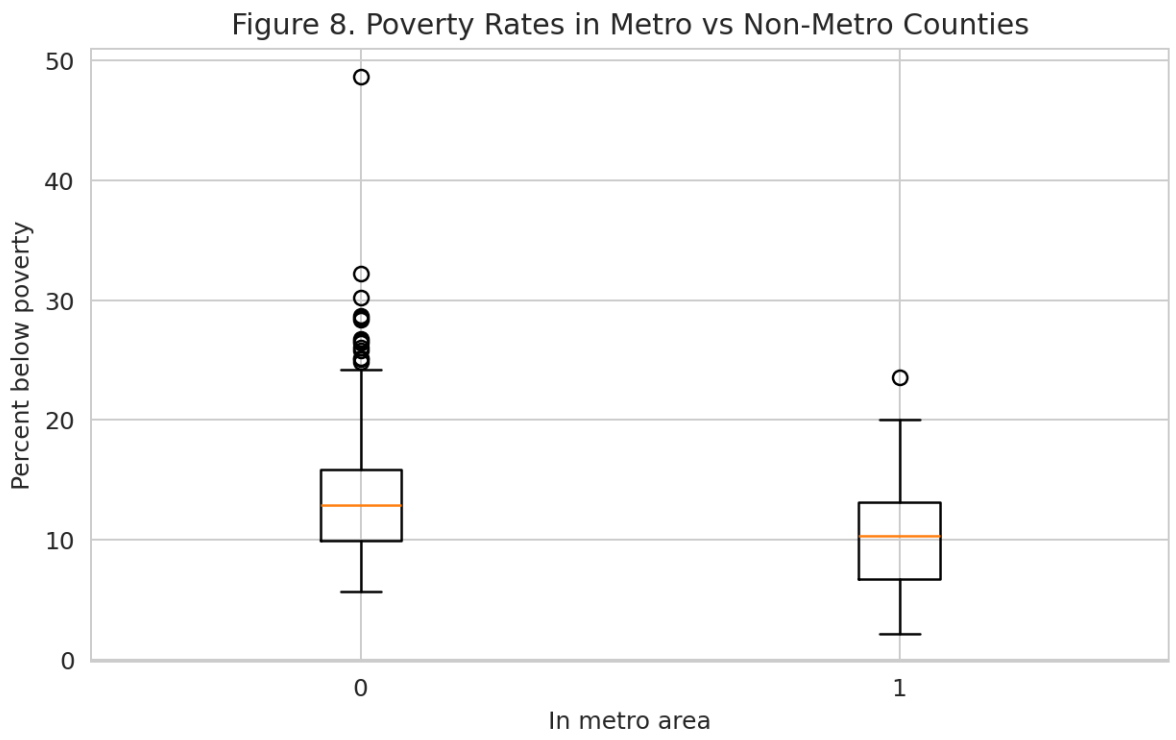

Figure 8: Poverty rates in metro versus non-metro counties. Non-metro counties have a higher average poverty rate.

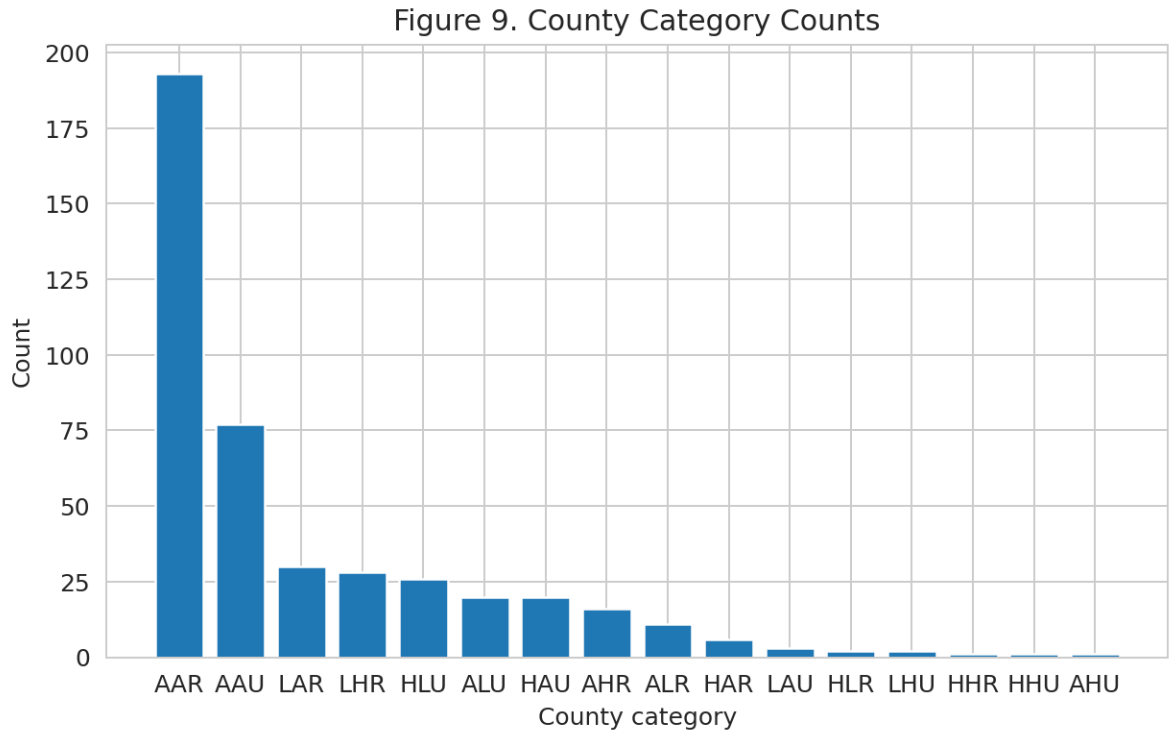

Figure 9: County category counts.

SUMMARY OF FINDINGS

This exploratory analysis shows that the Midwest dataset is rich enough to reveal several clear county-level patterns. Population-related variables are highly skewed because a small number of counties are much larger than the rest. Education and employment measures move together in a positive direction, suggesting that counties with larger college-educated shares also tend to have higher professional employment shares.

The categorical and graphical analysis also suggests a meaningful metro divide. Metro counties have lower average poverty rates than non-metro counties, while the correlation analysis shows that many demographic percentages move in predictable opposite directions. Overall, the dataset is well suited for EDA because it supports descriptive, relational, and visual analysis without requiring major cleaning.